

Consensus Analysis and Modeling of Visual Aesthetic Perception

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Abstract—This paper reports a characteristic relation between skewness and kurtosis of aesthetic score distributions in a massive photo aesthetics dataset generated from online voting. Analysis results reveal an unexpectedly wide range of kurtosis in the mediocre photo group, asymmetric consensus, the 4/3 power-law regime in both extremes, and tag-specific relation in the skewness-kurtosis plane. From the human cognition perspective on affective content analysis, these patterns are interpreted as supporting the necessity of a consensus property in addition to the preference used so far for accurate modeling of aesthetic evaluation process in human mind. For explaining the observed patterns, we propose a new computational model of a dynamic system based on the interaction between multiple attractors. Characteristic patterns in response time and consensus are predicted from the proposed model and observed in the experiments with human subjects for model validation.

Index Terms—Cognitive models, affect sensing and analysis, human information processing, perceptual reasoning



1 INTRODUCTION

STUDIES on visual aesthetics have gained an increasing attention during the last decade in the fields of psychology [1], neuroscience [2], and computer science [3], [4], concurrently having their own focuses. Data-driven analysis of image aesthetics has been proven as useful for elaborating the search result from massive photo collection [5] or suggesting the best scenic angle for amateur photographers [6], to name a few. Due to the nature of aesthetics as a highly subjective concept, like other topics in affective content analysis such as emotion and preference, an interdisciplinary approach has been regarded as essential for tackling the challenges.

From the view of affective computing [7], [8], while it is regarded as a part of emotion, aesthetics has a unique aspect that it is hard to specify benefit of the aesthetic appreciation, although a few researchers with Darwinian perspective suggested the benefit of recognizing several aesthetically pleasing and displeasing factors in preying or mating [9]. Scherer [10] pointed out, without aesthetic emotions, utilitarian emotions are not enough to explain affective response of human being.

In the field of psychology, and neuroscience recently, people have reported various “aesthetically pleasing” or “preferred” factors in images including color [11], [12], [13], [14], curved object [15], [16], contour [17], canonical size [18], [19], and spatial composition [20], [21] (see [22] for categorized summary). Compared with the factor analysis, relatively few conceptual models [23], [24] have been proposed to explain the effect of such factors to aesthetic perception.

In case of computational aesthetics, most researches have focused on recognizing the empirical heuristics of

painters and professional photographers such as composition [5], people and vanishing points [25], or rule of thirds [26], to name a few. In addition to the heuristic approaches, several researchers have shown the possibility that generic low-level features such as color histogram or GIST [27] can be effective for estimating the visual aesthetic value of a photo in the framework of machine learning [28], [29].

Throughout the various efforts of modeling visual aesthetic perception, one of the lasting issues is how to treat the mediocre samples, which receive 5 or 6 in a 10-point scale or 3 in a 5-point scale. Since Datta [29], it has been consistently reported that excluding the mediocre group from the training set significantly enhances the performance of the aesthetics estimator based on machine learning techniques [3], [30]. Also, the mediocre group causes two technical issues in machine learning approaches: imbalance problem and inappropriate sampling. The huge difference in size between the mediocre and the other groups, usually reaching 8 to 2 or even 9 to 1 in their ratio, makes most machine learning techniques suffer from bias to the mediocre in learning unless any appropriate resampling technique is used. However, rebalancing by resampling usually raises another issue that it is hard to choose most representative samples for mediocrity. That is why previous researches on computational aesthetics have excluded the ambiguous mediocre group at least in the training stage [30] or also in the test stage [29].

Considering the nature of aesthetic evaluation that most samples are evaluated as mediocre, we think that such exclusion might not help resolving the fundamental issue in the way toward a realistic model of aesthetic appreciation. Any two-class aesthetic value estimator trained solely from the two extreme groups might suffer from frequent false positives due to the majority of the mediocre in new incoming photos and thereby limit its application. Another machine learning alternative to such

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classification approaches is regarding it as regression problem [31],[32]. ACQUINE [31], the online photo aesthetic analysis engine, used the distance from the hyperplane in the SVM classifier as an aesthetic score ranging from 0 to 100. However, a large variance of scores was observed in the mediocre group, resulting in low correlation between ground truth and estimated scores [33]. Wu et al.[32] raised a more fundamental question on how representative a scalar value (e.g., an average score) is for aesthetic evaluation, claiming that a vector of score distribution should be used directly with a structural regression algorithm for capturing the nature of subjectivity.

This situation motivated us to analyze the properties of aesthetic evaluations, including the mediocrity, in large scale which is now available on the Internet owing to the popularity of social networks and photo collection services like Flickr. Specifically, as it is a matter of whether or not we are able to train a computational model from data, we were interested in the typicality of the good, the mediocre, and the bad; i.e., the consensus of rating for each group. The consensus of rating was previously mentioned by another research group [34]; they discarded the middle 80 percent of 60,000 photos in terms of average score while constructing their aesthetics dataset to make it learnable from the view of machine learning, based on their assumption that the mediocre ratings lack consensus among raters. However, their assumption regarding the average score as the metric of consensus needs to be validated further because such assumption requires another strong assumption that the degree of perceived beauty is identical among various factors: technically speaking, it does not discriminate between signal and its confidence. Another researcher group [32] also criticized the weakness and limitation of mean as an invalid measure.

Therefore, as a new method of measuring and visualizing consensus of aesthetic scores among people (and thereby typicality of the stimulus) for a myriad of photos efficiently, we propose to use a skewness-kurtosis map. Variance (the second moment, m_2), the popular choice for measuring consensus, is rejected because it is seriously distorted by highly skewed and bounded data. We regard kurtosis (the fourth moment, m_4) as a good alternative to variance because it indicates the lack of shoulders or infrequent extreme deviations [35]. For example, if almost all raters scored 5 points, its kurtosis would be significantly higher than the case that only a half of raters voted to the same point. Definitely, the interpretation of kurtosis should consider its skewness (the third moment, m_3) because there is a well-known relation of $K = S^2 + 1$ for Pearson's fourth moment we use. Under the assumption of unimodality, skewness can be regarded as a representative property of showing a major score in a group, regardless of its asymmetry which distort mean (the first moment, m_1). Therefore, combining the two properties, the skewness-kurtosis map can be a good visualization tool for revealing the degree of consensus for each score group.

While we believe we are the first who adopted the S-K map to aesthetics, the map has been used in other fields such as plasma physics, atmospheric science, oceanogra-

phy, or financial engineering where deviations from Gaussianity are investigated: See [36], [37] for brief review.

2 CONSENSUS ANALYSIS

2.1 Data Set

Owing to the Internet and digital photography, in the field of affective computing, crowdsourcing has become a popular method of gathering massive data of self-reported scores about visual stimuli via interactive environment such as Mechanical Turk (e.g., see [38]) or photography-dedicated web sites like dpchallenge.com. Among several available massive visual aesthetics datasets, our choice is the AVA (Aesthetic Visual Analysis) dataset [39] which has been publicly available since 2012. It consists of 255,530 photos and their 10-point (1 to 10) scores of aesthetics, rated by 200 (in average) photography professionals and hobbyists via online during a certain period of "challenge" in the website www.dpchallenge.com. During the rating, a photo was displayed in the web page with grey padding while preserving its original resolution and aspect ratio. The accumulated result by the previous raters was not exposed to a new rater.

A key benefit of the AVA dataset is, contrary to Photo.net [29] or CUHK dataset [34], that it preserves all score distributions, which is essential for consensus analysis. Another benefit is that it provides 65 textual tags (e.g., landscape, street, portraiture, food) describing the subjects or styles explicitly and thereby helps finding semantic factors in addition to consensus analysis. On average, 8,000 images are provided for each tag, and the mean distributions of aesthetic scores for tags are balanced: See Fig. 1 in [30] for the details. Another benefit of the AVA dataset is a large number of raters which enhance validity of higher moments; e.g., Photo.net [29] lacks raters for the purpose although it also provides massive photos.

2.2 Method

Because all observed samples are leptokurtic in the map, we use the excess kurtosis (the Pearson measure) for easy visualization and call it "kurtosis" through this paper. For all 255,530 photos in AVA dataset, we calculated quantiles and all four moments of score distribution for a photo and plotted the sample in the S-K plane; median and textual tags were additionally used to see group tendency.

In the S-K map, three reference trajectories were used to visualize the characteristics of the score distributions in comparative manner; Gaussian trajectory, Klaassen bound [40], and 4/3 power law trajectory. Gaussian trajectory depicts the parabolic relation of $K=S^2$ for unbounded Gaussian random samples. Klaassen bound is a theoretical lower bound of any unimodal distribution in S-K plane which is generated by the function $K = S^2 + 189/125$ [40]; for the purpose of unimodality check in the S-K map, the offset term of $-186/125$ in the original equation is adjusted to $189/125$ as a lower bound because we use excess kurtosis for K. This bound is usually drawn as a green solid line for all figures of S-K map in

this paper. The $4/3$ power law trajectory is generated from the function [37] of $K = 60^{1/3} S^{4/3}$. The last $4/3$ power-law regime, drawn as the red dashed line in this paper, was originally reported in the recent article [37] about the financial market data: the daily returns of S&P 500 stocks. While most physical data show parabolic near Gaussian region, a few the earthquake data and the stock market data are known to converge to the power law trajectory.

Considering the ubiquity of power-law relation in experimental psychology, it deserves to see whether or not the aesthetic evaluation, as one of various mental activities, shows the pattern which might signify a dynamic process which underlies the phenomenon.

Lastly, for the case where hypothesis test is required, Kruskal-Wallis test is used. Kruskal-Wallis test is one of the statistical non-parametric methods for comparing two or more samples by rank. More popular counterparts in parametric methods such as t-Test or one-way analysis of variance (ANOVA) are hardly applicable to the AVA dataset analysis because they assume a normal distribution of the residuals and therefore mislead to false interpretation when applied to the case that Gaussianity is not guaranteed.

3 ANALYSIS RESULT

Fig. 1 is the S-K maps for all 64 tags (the last tag 65 is excluded due to insufficient samples and visual convenience) which share the same axis of $[-2, 2]$ in m_3 (skewness, the 3rd moment) and $[0, 10]$ in m_4 (kurtosis, the 4th moment). Due to the characteristics of S-K plane, the “good” photos locate in the negative m_3 pane of the S-K map while the “bad” in the positive, which is inverse to the more familiar direction in raw score distribution

As a result, four patterns are observed in aesthetic score distributions from AVA dataset when projected to the S-K plane as following:

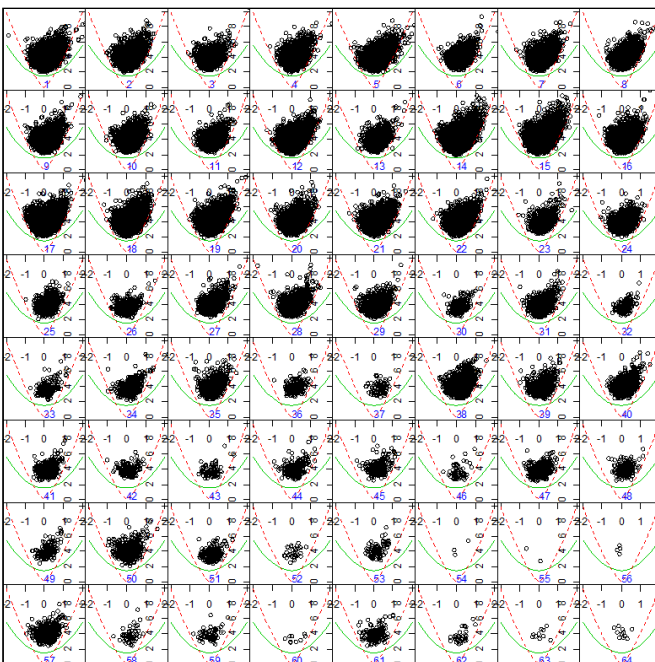


Fig. 1. The S-K maps of aesthetic score distribution for 64 tag groups

3.1 Pattern 1: a wide kurtosis range

Considering the nature of kurtosis as a consensus metric, the most notable pattern in Fig. 1 is the wide kurtosis range, from 2.0 to 10.0 approximately; even it reaches up to 8 for the symmetric samples (m_3 is near zero).

This clear non-Gaussianity is universal across all the tags to various degrees. Fig. 2 depicts the relation between skewness (m_3) and kurtosis (m_4) for the photos with the tag 14 (“landscape”) as one of the most representative tag groups. For visual clarity, they are clustered by median score (on top of each box) and the Gaussian parabolic relation $K=S^2$ (the bottommost blue dot line) is added to the other guidelines inherited from Fig. 1.

The bias of neutral point to the 6 median group is regarded as the side effect of the 10-point scale questionnaire that DPChallenge.com adopted because of the arithmetic middle point is not 5 but 5.5 in the scale as reported [41]. The proportionality between m_3 and m_4 in both extreme groups (2~3 and 8~9 in their median score) is a natural result of the mathematical relation between the two variables unless we consider its various scales specific to the tags.

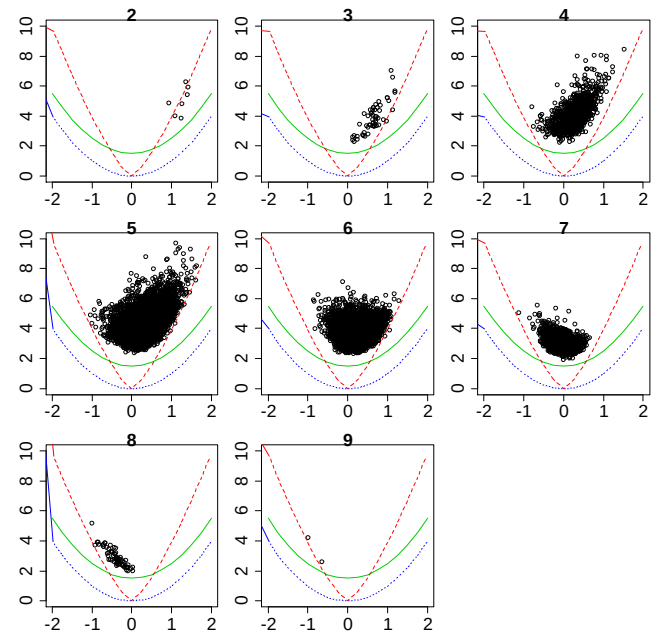


Fig. 2. Skewness-kurtosis relation of aesthetic score distributions for eight “landscape” photo groups clustered by median score from 2 to 9 (top of each plot): x-axis for m_3 (skewness) and y-axis for m_4 (kurtosis). Red dashed lines are power law trajectories, green solid lines are Klaassen bounds, and blue dotted lines are Gaussian parabolic relations.

Fig. 2 reveals that the wide range of kurtosis along the axis of $m_3 = 0$ in Fig. 1 is mainly caused by the mediocre group, not by the exceptional combination of the good and the bad groups. It means that the degree of consensus in the mediocre group varies greatly; in other word, the mediocrity originates from not only the “lack of consensus” but also the (almost) unanimous agreement.

For validating the interpretation on the wide kurtosis range, two contrastive samples are selected for comparing

the raw score distributions and their normality. Fig. 3 shows the score distributions and the Q-Q plots with normal distribution for two contrastive samples selected from the lowest ($m_4 = 2.44$) and the highest ($m_4 = 6.68$) kurtosis region respectively along the virtual axis of $m_3 =$

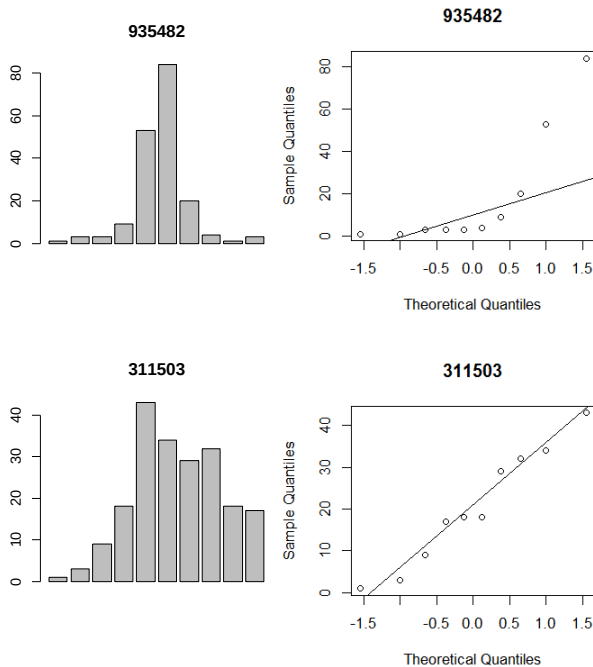


Fig. 3. A score distribution and a Q-Q plot for normality test for contrastive samples.

0 in the S-K map.

In Fig. 3, the upper row represents the sample (ID: 935482, number of raters = 181) at the point of $m_3 = -0.02$ and $m_4 = 6.68$ in the S-K map, and the lower row does the opposite sample (ID: 311503, number of raters = 252) at $m_3 = -0.01$ and $m_4 = 2.44$. As clearly depicted in the Q-Q plots between the observed distribution and the fitted normal distribution, the high m_4 sample failed in Shapiro-Wilk normality test by p -value = 0.0004 while the low one passed it by p -value = 0.7143 (all with 95% confidence intervals). Although all photos are excluded from this paper due to a potential copyright issue, the photos are freely accessible at the website www.dpchallenge.com: for an instance, the photo 935482 is accessible by using the URL www.dpchallenge.com/image.php?IMAGE_ID=935482.

3.2 Pattern 2: consensus asymmetry

Another property observed in Fig. 2 is the asymmetry of kurtosis range between the two contrastive groups; the m_4 of a “bad” sample tends to be higher than that of a “good” one. To measure the degree of asymmetry for each tag, the kurtosis distribution of two extreme groups are visualized by boxplot pairs of two extremes as depicted in Fig. 4: for reasons of space, only 33 tags are plotted.

To ignore the tag-specific preferential bias, two quartiles of mean score for each tag group are used as thresholds: $m_1 > Q_3$ (75th percentile) for the good group and $m_1 < Q_1$ (25th percentile) for the bad. Fig. 4 confirms that a

score distribution for a bad photo tends to have significantly higher kurtosis than a good one for all tags except several style tags: in Kruskal-Wallis test between the two groups for all tags, only the tag 54 (texture library), 55 (overlay), 60 (pinhole), and 62 (lensbaby) failed rejecting the null hypothesis with a 95% confidence interval. It can be interpreted as superiority of negative aesthetic evaluation to positive one, or vice versa, from the view of making consensus.

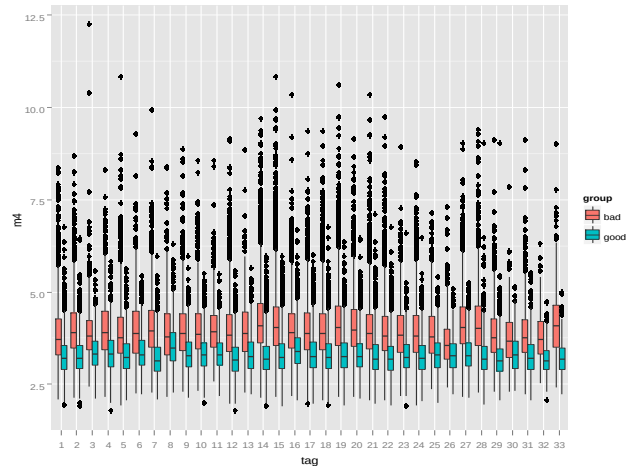


Fig. 4. Boxplot pairs of the m_4 distribution for 33 tags: for each pair,

3.3 Pattern 3: the 4/3 power law regime

Fig. 2 shows that, for “landscape” photos in AVA dataset, there is a positive correlation between the proximity of samples to the 4/3 power law trajectory in the S-K map and the score offset from the “neutral” point (5.5 or 6 in 10-point Likert scale).

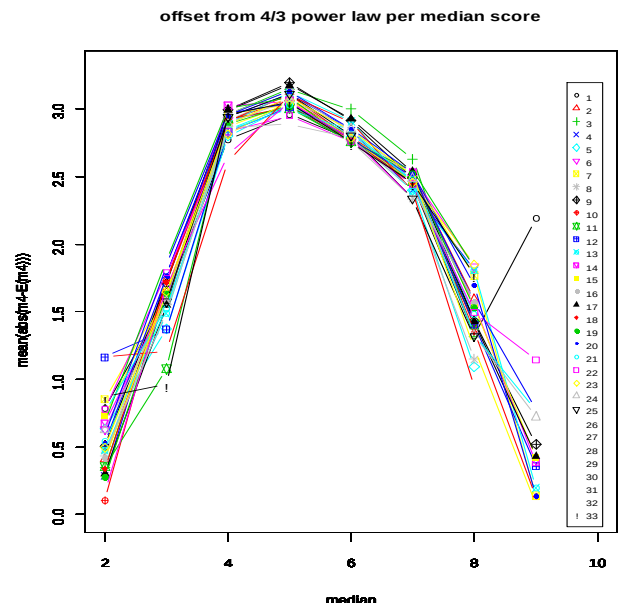


Fig. 5. Mean offset from the 4/3 power law per median score

For example, the most samples from the group of median score is 3 or 8 locate near the power law trajectories while samples from the neutral score group (median score is 6) are relatively scattered in the S-K map. Even though such

a convergence to the power law trajectories (red dashed lines) in both extreme groups (“very good” or “very bad”) can be partly explained as a truncated normal distribution which cause ceiling and flooring at the score of 1 and 10 respectively, a significant number of samples near the trajectories from the neutral score group raise an issue of strong non-Gaussian property behind it.

Fig. 5 illustrates the common tendency of the convergence across the first 33 tags by comparing the mean distances between observed kurtosis and the estimated value from the 4/3 the power law per median score. It also shows that inter-tag difference significantly increases for both extreme score groups.

Additionally, it is evident all samples locate above the unimodal distribution bound [40] of $K = S^2 + 189/125$ in Fig. 1 and Fig. 2.

3.4 Pattern 4: tag effect

While the above three patterns seems almost universal among the photos, the degrees of the patterns are affected by tags. For consensus, the Kruskal-Wallis rank sum test on the effect of tags to kurtosis, the consensus metric in this paper, rejected the null hypothesis of equality by p -value = $2.2e-16$ (with a 95% confidence interval).

Fig. 6 show each tag group in the scatter plot of the position of the offset from 4/3 power law and “m3_topm4,” the mean skewness of the samples in the top 25 percentile kurtosis.

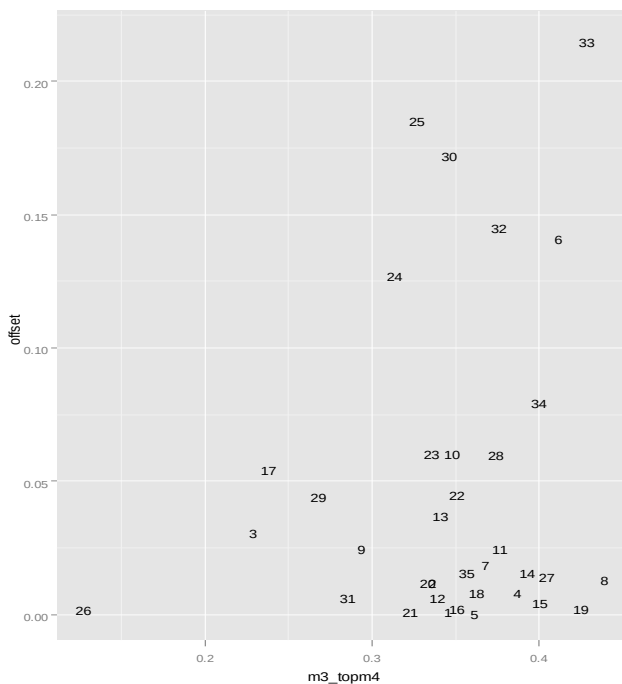


Fig. 6. Scatter plot of asymmetry and 4/3 power law offset

Therefore, a sample locates in the lower right pane of the plot if it is close to the power law regime and asymmetric. The distribution of tags in the plot shows a tendency that more asymmetric and power-law-compliant photo groups, usually located in the lower right quadrant, tend to be assigned with the tags about natural and spatial attributes including snapshot (“8”), animals (“19”),

landscape (“14”), nature (“15”), rural (“27”), sky (“7”), still life (“18”), and seascape (“35”), while the most symmetric or far-from-power-law group at the lower left or upper right quadrants tends to be followed by the tags about abstract concepts including action (“24”), photo-journalism (“25”), political (“30”), sports (“9”), and fashion (“3”). The mean amount of images for natural and abstract groups are 2287 and 518 respectively.

3.5 Discussion

Analysis in the higher moments, skewness and kurtosis, provides a new perspective to the aesthetics data interpretation. While Murray et al. [30] saw the aesthetic score distribution for a photo in the same AVA dataset as “largely Gaussian” after the review in the low moments regarding strong non-Gaussian characteristics in the both extremes as the floor-ceiling effect, we found that there is a strong non-Gaussianity (very likely to be a member of a power law family) in the same data showing a mixture of non-Gaussian and Gaussian distributions in the S-K plane. They also reported another important but misinterpreted property that the standard deviation of score distribution is proportional to the offset from the mean. We think the innate distortion of variance for highly skewed data and loss of information about “variance of variance” in the low moments are responsible for such insufficiency in the previous explanation.

Especially, we believe that kurtosis of score distribution is a good measure of consensus among raters and it should be treated as a key factor in modeling aesthetic evaluation process. The presence of a the new factor can explain why learnability of CUHK dataset [34] is so high, by interpreting it as the result of excluding all low consensus samples and thereby resampling tailored data only. In addition, the conceptual similarity between the concept of consensus and the confidence in signal detection theory deserves to be investigated further.

Another major pattern of asymmetry in kurtosis toward negative evaluation is in coordination with the lesson acquired in other researches on emotion. In the field of human computer interaction (HCI), there is a similar report that negative aesthetic decision on webpage design is made faster than the positive one [42]

For the third pattern, the regime of the 4/3 power law, Cristelli et al. [37] insist that it implies the presence of interaction between multiple agents behind the phenomenon. We agree with the opinion believing that visual aesthetic evaluation can be modeled in the similar manner, while treating the consensus issue separately.

For the fourth pattern of tag effect, the discriminative relation between natural objects and abstract concepts is in accordance with the previous studies which tried to explain the inborn preference in the framework of evolutionary psychology [43], [44]: especially, an innate preferential bias toward landscape has been studied thoroughly as the result of adaptation. Another qualitative report mentioned a similar but unclear tendency that a photo with more variance in its ratings is usually non-conventional [30]. Such an interpersonal similarity implies that consensus might be the matter of more “hard-

wired" attractors.

4 MODELLING

For comprehensive explanation on all the patterns we found, we have searched for previous models of visual aesthetic evaluation. From our knowledge, just a few researchers [23], [45] have proposed conceptual models of visual aesthetic evaluation. For example, Leder et al. [23] proposed an information-staging process model of art perception consisting of the five-stage modules in cascaded manner. Pelowski and Akiba [45] proposed a similar multi-stage model in the same domain. However, unfortunately, we have not found any previous aesthetic evaluation model which is able to explain the characteristic patterns we observed in the massive dataset quantitatively. While we and Wu et al. [32] share the same intuition that there is a sample-specific difference in consensus level, their genuine approach of multi-label classification regards these as given, not trying to model the underlying mechanism.

Therefore, we have devised several models for the purpose of acquiring a quantitative alternative. Motivated by the 4/3 power law regime and the exceptionally wide kurtosis range, our approach for the modelling is based on the dynamic process of multiplicative interaction between several positive and negative attractors with ambient Gaussian noise. In this scheme, the level of consensus and inter-tag difference in the S-K plane is determined by the tag-specific configuration of attractors. We expect that a successful model should be able to simulate the four patterns we observed in the AVA dataset.

For understanding the relationship between models and their characteristics from the view of the four-pattern representation, a static model with multiple attractors is analyzed firstly and then it expands to a dynamic model.

4.1 Static models

The simplest model would assume one aesthetic factor which varies in its degree of effect following a certain probabilistic distribution. For an instance, absolute difference between an imaginary ideal spatial layout and an observed one in an image, if exist, can be regarded as a single static factor of determining perceived aesthetic value, followed by random noise.

Fig. 7 shows the S-K plane projection of Monte-Carlo simulation results using the single-factor static model using Gaussian distribution for the mediocre group with three different standard deviation ranges, the very indicator of consensus in Gaussian distribution, while sharing a same mean of 5. In this simulation, three different ranges of standard deviation with eleven steps are used for Gaussian random number generation and number of trials for each configuration is determined as 200, following the average number of voters for each photo in AVA dataset. Considering the bounded nature of scores between 1 and 10 in the voting system used in AVA dataset, truncation in the same range of 1 and 10 as minimum and maximum is applied to the second ($sd = [1.0 \ 3.0]$) and the third ($sd = [3.0 \ 5.0]$) configuration; without truncation,

increasing variance does not affect the projected form in the S-K plane by showing as same patterns as the first configuration ($sd = [0.5 \ 1.0]$) in Fig. 7.

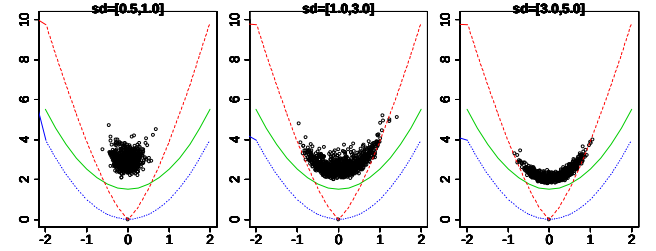


Fig. 7. S-K plots for Gaussian distributions with three different standard deviations.

Their scattered pattern in the S-K plane reveals that changing standard deviation in Gaussian distribution does not help simulating the wide kurtosis range and asymmetry we observed, although it simulates the 4/3 power law regime, mainly due to the truncation effect.

Considering the accumulated evidences about the plurality of aesthetic factors in the field of experimental psychology, a multiple factor model rather than the above single factor model is regarded as more appropriate to explain human visual perception of aesthetics. In case of color and brightness, since Eysenck[46], many researchers [11], [12], [13], [14], [47] have reported that there is a systematic pattern in group color preference: in hue preference, green, cyan, and blue are generally preferred to red and yellow [13], [47]; saturated colors are generally preferred [12], [14]; and, hue-value interaction exists in the way that a brighter image is generally preferred with different peak points for each color [11], [13]. In case of spatial structure, preference to horizontal and vertical lines [48], 1/f power spectra preference [49], [50], golden ratio [51], [52], symmetry [53], [54], soft curvature [15], [16], [55] and canonical composition [18], [19], [21], [56] have been reported as significant factors for visual aesthetic perception.

Such a multiple factor model can be implemented in various approaches. For an instance, the net value of perceived aesthetics can be a weighted sum of two or more factors; e.g., a spatial layout and a color tone. Our approach is clustering multiple factors as the two groups, positive and negative, and regarding these as two group factors. Among the several probabilistic distributions which treat two or more factors, beta distribution is selected. The probability density function (pdf) of beta distribution is a power function:

$$f(x; \alpha, \beta) = C \times (\alpha-1) (1-x)^{(\beta-1)} \quad (1)$$

where the normalization constant C is the product of three gamma functions $\Gamma(\alpha+\beta)$, $\Gamma(\alpha-1)$, and $\Gamma(\beta-1)$. From the view of aesthetics modeling, the simulated perception of visual aesthetics can be interpreted as the product of "good" (x) and "bad" ($1-x$) with their own powers. In this model, the two shape parameters, α and β , are regarded as the degree of effect that the two factor groups evoke in their respective directions; e.g., strength and numbers of

factors in a positive group determine the shape parameter α collectively.

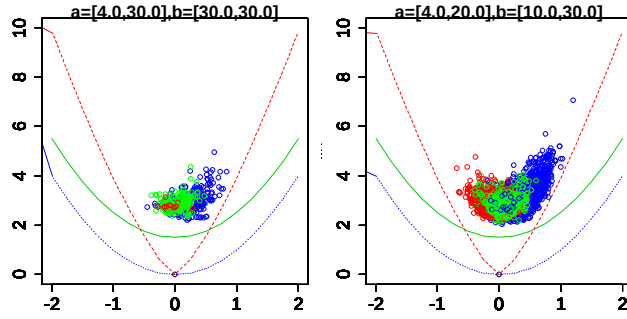


Fig. 8. S-K plots for two combinations of the α and the β ranges. Colored by their median aesthetic scores v : red for $v \geq 6$, blue for $v \leq 4$, and green for the others.

Fig. 8 is the simulation result of beta distributions generated from two different configurations. The left configuration in Fig. 8 changes the power of “good” (α) while keeping the power of “bad” (β) as constant. The right one changes both powers with structural bias to the dislike factors by higher maximum value of β . In comparison with Fig. 7, Fig. 8 shows a better result in that it meets all requirements except the wide K range in the mediocre group; especially, the second pattern, consensus asymmetry, is easily represented by rebalancing the power ranges of the two factor groups.

Although we used the simple beta distribution model with two factors for clarity of explanation, if one of the two factors consists of multiple subfactors, this model can be easily extended to Dirichlet distribution, the multivariate generalized version of beta distribution.

4.2 Dynamic models

The last unresolved issue of the wide kurtosis range in the static models for aesthetic evaluation reveals that we need another computational model which is able to produce significantly different evaluation results in accordance with small variance in model parameters among people. Also, it seems desirable to inherit the concept of two contrastive multiple factor groups that we showed in the previous beta distribution model as it is useful for simulating consensus asymmetry.

At this point, we consider dynamic systems as the candidate for revising the model based on our knowledge that it is easy to observe such a parameter-sensitive change (e.g., bifurcation) with the models of dynamic systems [57]. Among various dynamic models, a Drift-Diffusion Model (DDM) [58], [59], [60] has been popular among psychologists as a well-defined model of explaining behavioral data of reaction time and accuracy for the task of forced categorization among two or more alternatives (See Ditterich [61] or Bogacz et al. [59] for review). This model assumes that human mind requiring a binary decision (A or B) accumulates evidence favoring each alternative over time, while simultaneously distracted by internal random walk (noise), and makes decision when the accumulated evidence for either alternative is enough.

This process is usually depicted as a particle drifts and diffuses between the two boundaries until it reaches either boundary. In this view, each boundary attracts the particle at a given or varying rate with noisy turbulence, making the process stochastic. For an instance, in a traditional DDM assuming only one attractor in one side, the evidence is accumulated in according to

$$dx = A dt + W, \quad x(0) = 0. \quad (2)$$

In Equation 2, x grows at rate A in average while white noise (W) is continuously added [59].

If we regard aesthetic perception as a compromise between “good” and “bad” factors which attract in opposite directions, the DDM can be used to model the dynamic process with the variance in the number and positions of the attractors in both sides.

From our knowledge, temporal properties of affect models have not been fully explored except a few models like WASABI [62] or componential appraisal theory [10]. For modeling the interaction between multiple attractors with moderate randomness, motivated from the diffusion decision model [60], we propose a new model named Drift Diffusion Model for Aesthetic Perception (DDM4AP). Because aesthetic perception is usually captured as a multi-label classification problem using n -point Likert scale (e.g., 10-point scale in AVA dataset), it is inevitable for the new model to modify the original drift-diffusion model, which assume 2AFC, significantly except its core concepts, although aesthetic perception can also be approximated to the binary decision of 2AFC: like or dislike.

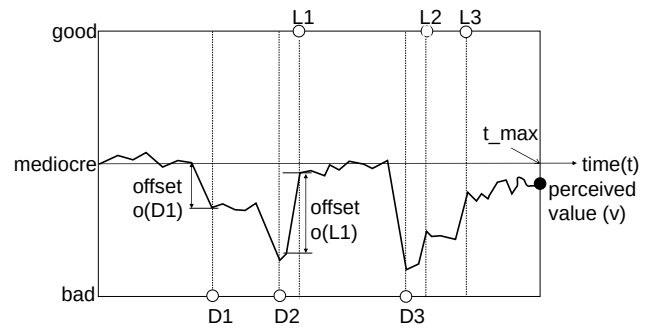


Fig. 9. An example of DDM4AP with three “like” factors at the good side (top) and “dislike” factors at the bad side (bottom) respectively.

Fig. 9 shows an illustrative example of DDM4AP having equal numbers of attractors in the both sides, the good and the bad. The perception process ends when the reference point (the black dot in Fig. 9) reaches the end of time axis, $t_{\max}$, or either border (top or bottom) before $t_{\max}$. The finally perceived aesthetic value (v) is mapped from 1 to 10 along the three borders starting from the bottom (“bad”) through the rightmost border to the top (“good”); as the 10-point score follows a Likert scale, the score distribution along the borders can be arbitrary as far as it preserves the order. If there are three L attractors and one D attractor, the landing position of the

reference point will be systematically biased to be somewhere on the top or rightmost border while the offset is determined by the drift rate; if there isn't any attractor in the stimuli, the perceived value of DDM4AP is solely determined by diffusion, which is usually modeled as bounded white noise.

In DDM4AP, two factor groups, L (Like) at the top side and D (Dislike) at the bottom, are treated as the collection of attractors: L_i for good and D_j for bad respectively. The number and the sequence of attractors along time axis at both sides significantly affect the resultant aesthetic value at the end of the process. In Fig. 9, offset $o(D_1)$ depends on the probabilistic gain of the first attractor D_1 while offset $o(L_1)$ is determined by L_1 located on the opposite side.

Therefore, at the end of the process, the perceived aesthetic value v is determined as following:

$$v = \text{neutral score} + \sum_{i=1}^m o(L_i) - \sum_{j=1}^n o(D_j) + W \quad (3)$$

where m and n are the number of attractors in the good and the bad side respectively which exist before the end position at time axis, W as a uniformly distributed net diffusion, and $o(L_i)$ or $o(D_i)$ are random variables following Gaussian or exponential probability density function (pdf). The neutral score in Equation 3 is usually set to 5 or 5.5 in the 10-point scoring system.

Due to the characteristics of drift-diffusion system, the temporal distribution of attractors is also an important factor affecting not only latency but also the offset from neutral evaluation. For example, the perceived value (v) in Fig. 9 would be systematically biased to "bad" if the D group locates prior to the L group. The idea of temporal distribution of attractors is justified by the various processing times in all levels of perception and cognition.

Because we assume that the position and gain of the attractors are different photo by photo, the simulation results are generated by overlapping the responses from 200 people for 100 photos.

In our Monte-Carlo simulation, the positions of attractors are determined by uniform random number generation between 100 and 300 trials while their gains follow the exponential distribution with $\lambda = 5.0$ or Gaussian distribution. The noise term is implemented as uniform random distribution (white noise) with the gain of 0.015.

Fig. 10 and 11 show the simulation results of DDM4AP with two different types of attractors (Gaussian and exponential) which meet all requirements we proposed. For each m:n case, the fourth quadrant shows a simplified DDM4AP trajectories down to ten raters with the position marks of L_i (the green semi-circles in the upper bound) and D_i (the red semi-circles on the lower bound). Contrary to the static model, it simulates the wide K range for

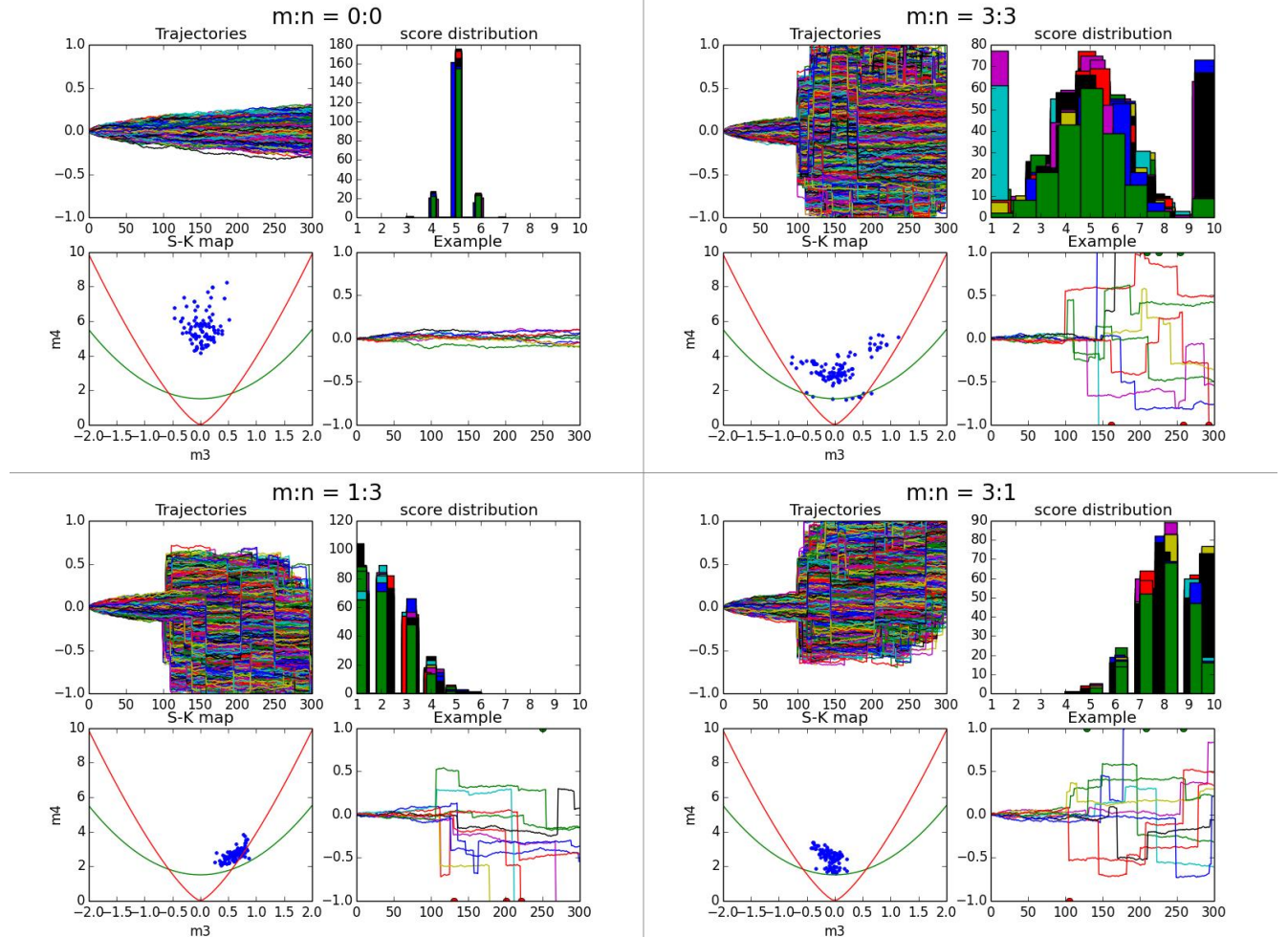


Fig. 10. Simulation results of DDM4AP with Gaussian attractors.

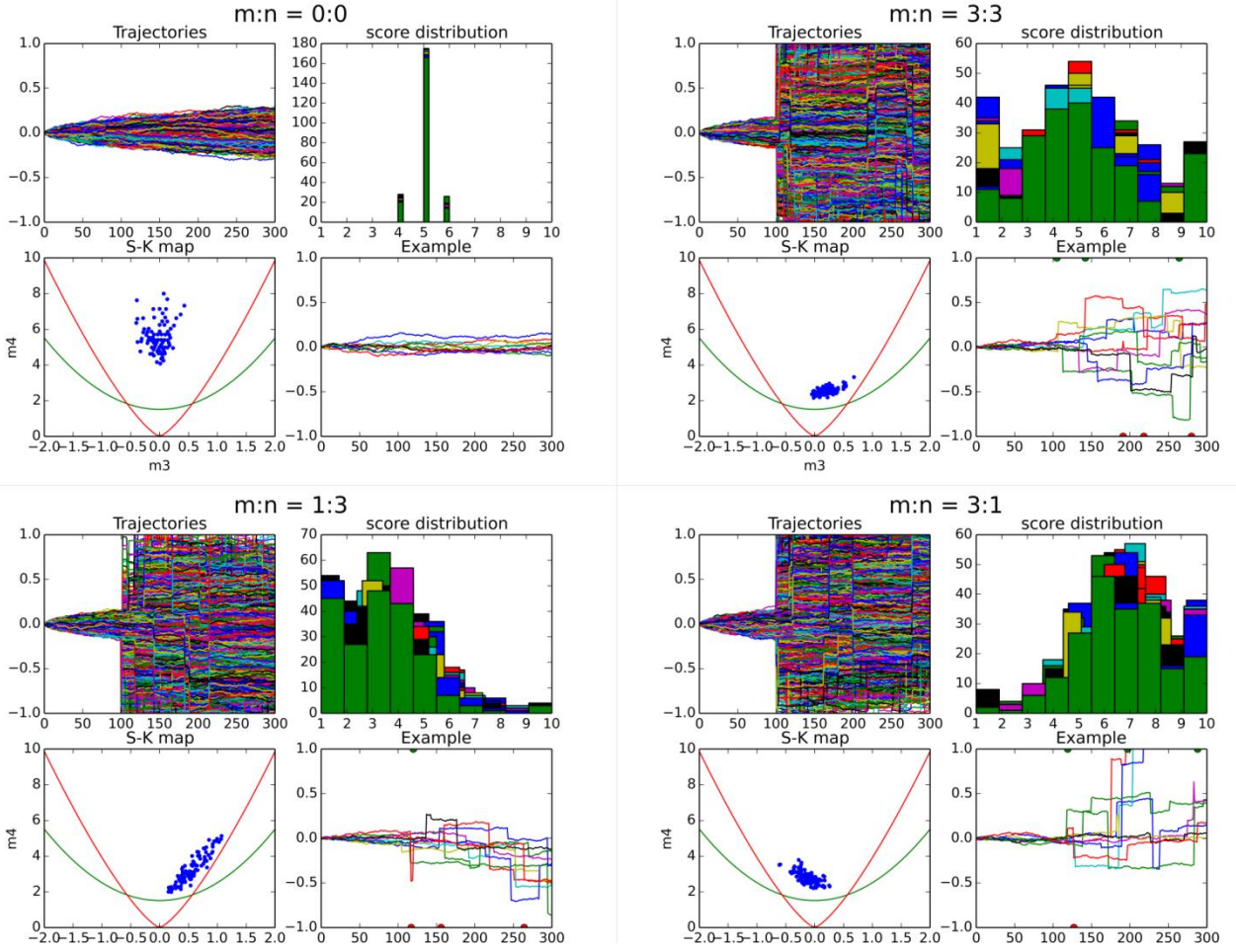


Fig.11. Simulation results of DDM4AP with exponential attractors

the mediocre group (L and D are balanced or equally void) in the S-K map while mimicking asymmetry by rebalancing drift between the two attractor groups. Comparing the two figures, assuming exponential attractors results in more realistic score distributions rather than its Gaussian counterpart; contrary to the exponential version in Fig. 11, the Gaussian version in Fig. 10 produces too biased score distributions in our setting.

At least in the same setting, a specific case of 0:0 in the ratio of attractor numbers between the two groups is responsible for such a high kurtosis in the mediocre group, given that the diffusion rate is small enough compared with drift rates. This result is interpreted as natural in DDM4AP because the other balanced case ($m = n$) might generate more various results due to it is apt to be affected by small difference in attractor gain and process time (therefore the position in time axis) for a common factor among people.

The following Table 1 summarizes the four tested models from the view of accordance with the four observed patterns we found in AVA dataset and one additional rule (Klaassen bound). The table reveals the relatively high potential of dynamic system models rather than static models for explaining all the observed patterns.

As shown in the simulation results in Fig. 10 and Fig. 11, DDM4AP leads to several predictions about response

time (or latency) from the view of a dynamic system as following:

TABLE 1
COMPARISON BETWEEN MODELS

Requirement	Gaussian	Beta	DDM-G	DDM-E
Convergence to power law in extremes	Pass	Pass	Pass	Pass
Within Klaassen bound	Pass	Pass	Pass	Pass
Tag-specific effect	Fail	Pass	Pass	Pass
Consensus asymmetry	Fail	Pass	Pass	Pass
Wide kurtosis range	Fail	Fail	Pass	Pass

*DDM-G is DDM4AP with Gaussian pdf while DDM-E with exponential pdf.

First, the response times of aesthetic evaluation will be significantly different between score groups; specifically, the mediocre and the other (the good or the bad); in other word, evaluating the mediocre should take longer than the good or the bad. This phenomenon was previously reported in another domain, web design appreciation [42].

Second, the response time will be significantly affected by the kurtosis of rating distribution; e.g., if diffusion is

small enough compared with drift toward an either side, one stimulus with high kurtosis in its rating distribution is expected to have a longer response time than another stimulus with relatively low kurtosis.

5 MODEL VALIDATION

To validate the hypotheses induced from DDM4AP about response time, an experiment was conducted to human subjects in the following setup.

5.1 Stimuli

For concentrating on response time, tag effect was controlled by collecting stimuli from single tag group. Considering the relatively less-individual preference on real-world images [43], among the 3564 AVA photos that its tag1 is 14(“landscape”) and the number of raters is larger than 100, we selected 100 photos and classified into three groups - good, mediocre, and bad - according to the following criteria:

good if median score ≥ 7 ;
bad if median score ≤ 4 ; and,
mediocre if median score =5 or 6.

In case of the mediocre group, due to its relatively huge amount (3124 mediocre vs. 216 good vs 204 bad), it is filtered again by the mean and skewness of scores as following:

$5.0 \leq \text{mean score} \leq 5.5$; and,
 $-0.1 < \text{skewness of scores} < 0.1$.

Finally, for each group, the topmost and bottommost photos were selected from the view of score kurtosis: the top15 and the bottom 15 for the good and the bad, and the top 20 and the bottom 20 for the mediocre.

Conclusively, 100 photos were selected as stimuli consisting of three groups in accordance with the combination of median scores and kurtosis of score distribution (therefore the degree of consensus).

5.2 Subject

10 male and 15 female students in the age of 20s and 30s with normal or corrected vision participated in this experiment voluntarily with a small compensation of ten dollars for each person. By the activity level of digital photographing, the subjects are categorized into three groups (daily, weekly, monthly) consisting of 6, 18, and 1, respectively.

5.3 Method

To simulate the online rating environment in DPChallenge.com, a subject sat in front of a 24 inch LCD monitor which displays a photo (stimulus), shown at the center of the screen in original resolution with gray padding. The subject was requested to evaluate the aesthetic value of the photo on the screen in the same scale with DPChallenge.com by selecting a score button among 10 choices (from 1 to 10). Once the button was pressed, it proceeded to the next photo and repeated the same task until all one hundred stimuli in a random order were scored. Revisiting the previous photos was not allowed. During the

process, the selected aesthetic score and response time were recorded synchronously while the subjects were not aware of the recording of response time, under the control of PsychoPy software [63].

5.4 Experiment Result

Considering the inter-personal difference in the range of response time, the relation between response time and score was firstly investigated individually for each subject as shown in Fig. 12. In Fig. 12, a subplot with triangle marks represents a subject whose response times are significantly affected by score (with a 95% confidence interval): out of 25 subjects, 11 subjects (44 percent) were classified as significant. However, for each rater, the effect of the average response time was not significant to the score.

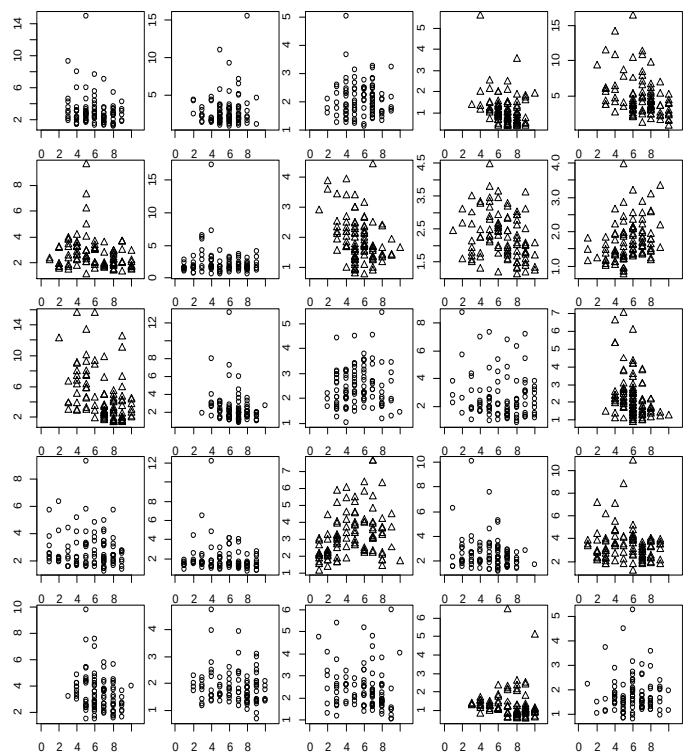


Fig.12. Latency distributions (y-axis in second) per score (x-axis) for 25 subjects: a triangle mark for the subjects whose response time is significantly affected by score, while a circle mark for the other (with a 95% significance interval).

For analyzing the general relation between response time and aesthetic score across subjects, quantile normalization was applied to adjust interpersonal difference in response time because of high non-Gaussianity including several outliers which are suspected as the result of temporary attention failure. Then, Kruskal-Wallis rank sum test was applied to see whether or not score affects response time as predicted, followed by the result of $p\text{-value} = 4.279\text{e-}15$ saying that score significantly affect response time with a 95% significance interval. With the same setting, another prediction of significant effect of kurtosis of scores to response time was also confirmed by $p\text{-value} = 4.335\text{e-}14$.

Fig. 13 depicts the relation between latency (response time) and aesthetic score. As predicted, the mean re-

sponse time is longer in the mediocre than the other groups. The second pattern, asymmetry between the good and the bad, is observed in response time comparison.

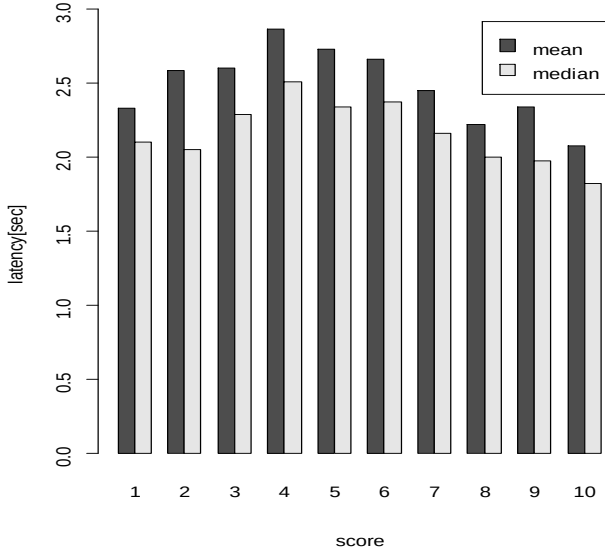


Fig. 13. Response times as a function of aesthetic scores

Table 2 is the result of pairwise post-hoc comparison using Wilcoxon-Mann-Whitney rank sum test with adjusted by Bonferroni's method: In this test, the confidence interval of 0.95 was adjusted to 0.967, which support the above validation in more quantitative manner. In the pairwise three-class comparison, there is a significant difference between the mediocre and the good (p -value = 0.00011) while the difference between the mediocre and the bad is not significant (p -value = 0.06826).

Table 2

Pairwise post-hoc comparisons (Wilcoxon-Mann-Whitney rank sum test with a "not equal" alternative hypothesis) between response times of three groups

	Bad	Mediocre	Good
Bad	N/A	0.06826	1e-05
Mediocre	0.06826	N/A	0.00011
Good	1e-05	0.00011	N/A

While latency was significantly affected by the kurtosis of scores for each stimulus (p -value = $4.335e-14$ with a 95% confidence interval), kurtosis of scores for the mediocre stimuli having 5 as their median score was not significant for affecting latency (p -value = 0.3624). We regard it as supporting the second prediction of DDM4AP as a persuasive model for visual aesthetic evaluation because, in the frame of DDM4AP, this result can be explained as the "timeout" for a mediocre stimulus is determined when a particle reaches the rightmost bound, not by the net drift time of the particle during the evaluation process.

In additional factor analysis, gender is proven as a significant factor for explaining both response time and

score (p -value = 0.000136 and $2.904e-12$ respectively). Fig. 14 shows the latency distribution of male and female subjects. The effect of mediocrity to latency is stronger in the female group than in the male group, while the male group shows more dispersed scoring pattern. Although such gender effect can be explained and simulated by difference in the confidence level, numbers and positions of attractors, and attractor gains in DDM4AP, we need more elaborated experiment design with more subjects to find out a key factor which is responsible for the gender difference.

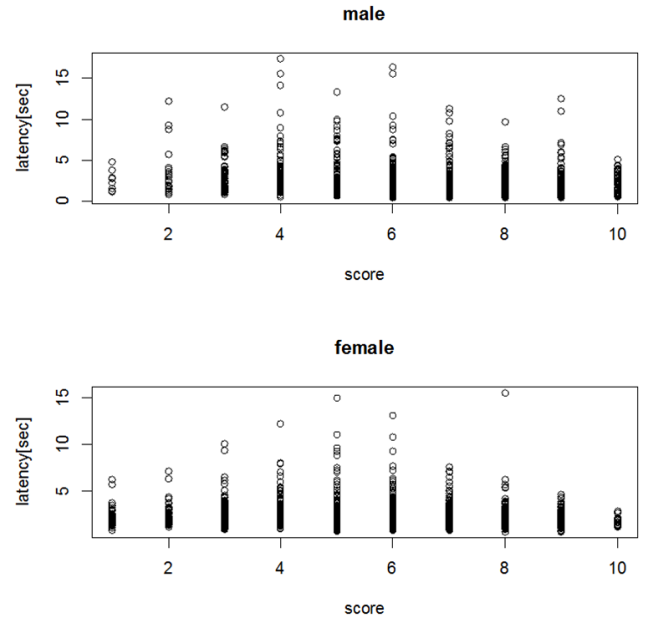


Fig. 14. Latency distributions from two gender groups

Lastly, we have analyzed the correlation between score and the activity level of photographing, as our subject group consists of six "daily" photographers, eighteen "weekly" and one "monthly": we asked the question, "How often do you photograph? Daily, weekly, or monthly?," while collecting participants via online. We analyzed the effect of the two activity levels, the daily and the weekly (the monthly is ignored due to too small sample size), to score distribution of each photo and counted how many photos show significant difference in score distribution between the daily and the weekly groups. Kruskal-Wallis rank sum test showed that, with the 95% confidence interval, only 10 out of 100 photos were significantly affected by the activity level in their score distribution; with the 99% confidence interval, only 2 out of 100 were significant. For the issue that our discovery might be biased to the group of "hobbyists and professionals" [39], even though we need an additional human subject test with massive number of professionals for complete validation as a future work, currently we estimate from the within-amateur analysis that the patterns we found will be consistent with various degrees from amateurs through professionals.

6 DISCUSSION

The experimental results support the two hypotheses from DDM4AP that response time is significantly affected by score group and kurtosis, the degree of consensus. The former one is previously reported in the research with the stimuli of web page design [42] with a few differences: in our experiments, the response time for the bad photos is longer than the good. We regard it is due to the difference in questionnaire design and culture, requesting further experiments under more controlled environment for valid comparison including a comparative study.

For the individual analysis result, the type of raters might affect score distributions among raters. In other word, if a rater's scoring is deviated from the average rating pattern or distorted by outliers due to the lack of attention, the correlation between response time and score is also distorted. It can be resolved by increasing the number of raters for each stimulus to the level of hundreds as AVA dataset provides.

One important aspect of DDM4AP is that it is able to simulate the translation between discrete emotion models and dimensional models; for an instance, valence can be interpreted as the result of dynamic interaction between several discrete emotion attractors as DDM4AP successfully visualizes.

The difference between mean and median response time implies there are two different mechanisms behind the process. In the frame of DDM4AP, drift-oriented decision and diffusion-oriented decision can explain the duality.

Another issue DDM4AP raises is that previous "stationary" machine learning models might be limited in their ability of simulating visual aesthetic perception (and, further, emotion), because they don't consider the properties of dynamic systems. If the visual aesthetic perception is the result of a dynamic system including multiple attractors as we modeled in DDM4AP, the mixture of positive and negative attractors in a training sample would misguide most machine learning methods unless they have a reliable active learning scheme, another difficult issue in the field of machine learning. It is due to the common assumption of one-to-one or many-to-one relationships, with noisy variance, between samples and their classes behind most machine learning methods, while DDM4AP allows one-to-many relationship additionally and regards "variance", or even "bifurcation", as a systematic consequence. The pioneering work of Wu et al. [32] using SVRD has potential of making synergic with DDM4AP. For an instance, DDM4AP might help SVRD treat samples with null stimuli, which include none of attractor, differently in the mediocre group, or adjust weights of score bands during training based on the level of consensus. Although SVRD didn't show a comparable performance in 9,000 photos from dpchallenge.com (the fraction of AVA dataset named as "DS2" in [32]) to the good result with Photo.Net ("DS1" in [32]), we believe that their approach of multi-label classification is fundamentally correct and promising, hoping to expand Wu's work in the future work of application: that will be a combination of a dynamic model in psychology and multi-label (1-to-N) classification in machine learning.

This explains why the pragmatic solution of excluding

low consensus samples during the training stage [34] was so effective to enhance the performance of traditional classifiers such as support vector machines or Bayesian classifiers in the task of aesthetics evaluation from low-level features. In the same vein, the usual practice of excluding images in the mediocre group during training should be also effective because the stimuli are the most likely to have multiple and balanced (between the positive and the negative) attractors, or nothing as a null stimulus; either case is sufficient to prevent classifiers from being learned.

From the view of computer vision, the predicted presence of two types of the mediocrity in DDM4AP implies we need to treat them differently to each other when finding features or training classifiers. For example, if a set of one-vs-one classifiers are used for multi-label classification, one mediocre sample without any like or dislike factor should be treated differently to the other mediocre sample having equal numbers of like-dislike factors. In other case, it would be more promising to construct a multi-label classifier from a set of one-vs-all classifiers if we can detect and exclude the mediocre sample of "balanced-between-two-attractor-groups" before training, because it is unlikely that the two types share a similar embedding pattern in feature space.

We hope that our approach helps giving intuition for breaking the "glass-ceiling" [64], which has been regarded as the consequence of the semantic gap between low-level features and high-level perception, in prediction of emotion. For an instance, it might deserves to try an active learning model for evaluating visual aesthetics of photos by adjusting learning rate or changing combination in ensemble methods according to the result of sophisticated consensus analysis on the rating pattern.

7 CONCLUSION

Statistical consensus analysis using higher moments of self-reported aesthetic evaluation data on a massive photo dataset reveals four characteristic patterns: a wide kurtosis range, consensus asymmetry, the 4/3 power law regime at both extremes, and tag effects.

Because a simple probabilistic distribution model (e.g., a unimodal Gaussian distribution) is inadequate to explain or simulate these patterns, several alternative models of visual aesthetic perception are proposed and evaluated by the presence of the observed patterns in their simulation results, concluding that a dynamic model named DDM4AP, a modification of drift-diffusion model, is most successful for simulating all the patterns owing to its mechanism of determining the perceived aesthetic value by the spatiotemporal interaction between multiple attractors with random noise.

To evaluate the feasibility of DDM4AP as a model of visual aesthetic perception in human mind, its innate property of dependency between perceived aesthetic values and their response times is tested via a human subject experiment. The experimental results show that

the dependency exists as DDM4AP predicts, supporting the model as reflecting core properties of visual aesthetic evaluation process.

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